

DATA ANALYST PORTFOLIO PROJECT

COVID-19 Mortality Analysis

Using Excel, SQL and Power BI

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Health Data Analyst | Medical Analytics and Informatics | Public Health Reporting

Area	Details
Project type	Health data analytics case study
Dataset	WHO COVID-19 global data file used for this portfolio project
Tools demonstrated	Excel, SQL, Power BI, DAX, dashboard design and health data interpretation
Date range analysed	2020-01-05 to 2026-04-26
Records analysed	79,200 rows across 240 countries/locations
Portfolio role alignment	Junior Data Analyst / Health Data Analyst / Health Informatics Analyst

1. Executive Summary

This project analyses reported COVID-19 cases and deaths using a public health dataset in order to identify mortality trends, compare countries and regions, calculate case fatality rates, and present the findings in an interactive Power BI dashboard.

The project demonstrates a practical end-to-end data analyst workflow: importing and preparing data in Excel, querying and validating metrics with SQL, building a data model and DAX measures in Power BI, and translating health data into clear insights for decision-making.

Portfolio positioning

This project is aligned to health analytics, medical informatics, monitoring and evaluation, public health reporting, and business intelligence roles. It is especially suitable for showing entry-level to junior data analyst capability in the health sector.

2. Candidate and Health-Sector Fit

Rumbidzai Mahlanganise has a health informatics background and is positioning the portfolio around health-sector analytics. The case study is therefore written to show practical ability in health data interpretation, dashboard reporting, and evidence-based decision support.

CV evidence area	How this project supports it
Medical Analytics and Informatics	The project uses a public health mortality dataset and converts it into clear decision-support outputs.
Excel, SQL, Power BI and Tableau exposure	The project focuses on Excel, SQL and Power BI as the first portfolio project, with space to extend into Tableau later.
DHIS2, EHR/Impilo and REDCap awareness	The analysis style is relevant to routine health reporting, public health indicators and digital health data quality.
Health Informatics Intern experience	The dashboard mirrors real reporting tasks such as system usage analytics, monitoring, reporting and decision-support insights.

3. Problem Statement

COVID-19 created a major need for accurate, understandable and regularly updated public health reporting. Health teams, policy makers and operational decision-makers needed to monitor cases, deaths, mortality trends and country-level comparisons in order to understand disease burden and respond effectively.

This project focuses on analysing reported COVID-19 deaths and cases to identify trends, compare countries, understand Zimbabwe-specific patterns and communicate insights through a clean interactive dashboard.

4. Project Objectives

- Analyse reported COVID-19 cases and deaths over time.
- Compare countries and WHO regions using deaths, cases and case fatality rate.
- Build a focused Zimbabwe analysis page to demonstrate national-level public health reporting.
- Create an interactive Power BI dashboard with filters for reporting year, country and WHO region.
- Communicate insights, limitations and recommendations in a clear portfolio-ready format.

5. Key Questions Answered

1. What were the reported global COVID-19 case and death totals in the dataset?
2. Which countries recorded the highest reported COVID-19 deaths?
3. How did reported deaths change over time?
4. Which countries had the highest case fatality rates when a meaningful case threshold is applied?
5. How did Zimbabwe's reported COVID-19 cases, deaths and mortality trends develop over time?
6. What limitations should be considered when interpreting reported COVID-19 data?

6. Tools Used

Tool	Use in project
Microsoft Excel	Initial data review, cleaning checks, pivot-style summaries and workbook preparation.
SQL	Structured analysis queries, aggregation, country comparison and validation of key metrics.
Power BI	Dashboard development, data modelling, DAX measures, slicers and visual storytelling.
DAX	Measures for total cases, total deaths, case fatality rate and latest reporting date.
WHO COVID-19 data	Source dataset for reported COVID-19 cases and deaths.

7. Dataset Summary

Dataset field	Portfolio value
Date_reported	Allows time-series trend analysis and weekly/monthly reporting.
Country_code and Country	Supports country-level comparison and filtering.
WHO_region	Supports regional comparison and slicers.
New_cases and Cumulative_cases	Supports case trend analysis and cumulative burden context.
New_deaths and Cumulative_deaths	Supports mortality trend analysis and death-burden reporting.
Metric	Value from dataset
Date range analysed	2020-01-05 to 2026-04-26
Rows analysed	79,200
Countries/locations	240
WHO regions present	AFRO, AMRO, EMRO, EURO, OTHER, SEARO, WPRO
Global reported cases	779,213,618
Global reported deaths	7,114,800
Global case fatality rate	0.91%
Zimbabwe reported cases	266,436
Zimbabwe reported deaths	5,740
Zimbabwe case fatality rate	2.15%

8. Methodology

Stage	Work performed	Output
1. Data collection	Collected the WHO COVID-19 global dataset and reviewed available fields.	Source CSV file.
2. Data preparation	Checked date fields, country fields, missing values and numeric columns.	Analysis-ready dataset.
3. Excel analysis	Created workbook structure, summary tables and dashboard-ready tables.	Excel workbook and summaries.
4. SQL analysis	Built queries for totals, trends, top countries, case fatality rate	SQL scripts and validation logic.

	and Zimbabwe focus.	
5. Power BI modelling	Loaded fact and dimension tables, created relationships and DAX measures.	Interactive data model.
6. Dashboard design	Built executive overview, country comparison, country details, Zimbabwe focus and insights pages.	Power BI portfolio dashboard.
7. Interpretation	Converted visual outputs into findings, limitations and recommendations.	Final case study and README.

9. Core Calculations and DAX Measures

The main dashboard uses additive weekly fields for total cases and deaths. Cumulative fields are useful for checking final values, but summing cumulative values across weeks would overstate totals because cumulative values repeat over time.

Measure	Purpose	DAX / logic
Total Cases	Total reported cases in the selected filter context.	SUM(Fact_COVID19_Weekly_WHO[New_cases])
Total Deaths	Total reported deaths in the selected filter context.	SUM(Fact_COVID19_Weekly_WHO[New_deaths])
New Cases	Weekly new cases for trend visuals.	SUM(Fact_COVID19_Weekly_WHO[New_cases])
New Deaths	Weekly new deaths for mortality trend visuals.	SUM(Fact_COVID19_Weekly_WHO[New_deaths])
Case Fatality Rate	Reported deaths as a proportion of reported cases.	DIVIDE([Total Deaths], [Total Cases], 0)
Latest Reporting Date	Most recent date in the filtered dataset.	MAX(Fact_COVID19_Weekly_WHO[Date_reported])

10. Power BI Dashboard Pages

Page	Purpose	Main visuals
Executive Overview	Provides the global summary and high-level mortality view.	KPI cards, trend charts, regional summary.
Country Comparison	Allows users to compare countries dynamically by year, WHO region and country.	Slicers, KPI cards, top 10 deaths, top fatality rate, death trend.
Country Details	Provides a detailed country-level table for checking exact values.	Country table with cases, deaths, fatality rate and latest date.
Zimbabwe Focus	Provides a focused national-level view for Zimbabwe.	Zimbabwe KPI cards, weekly trend charts, monthly deaths, insight boxes.
Insights & Data Limitations	Explains findings, public health meaning, limitations and recommendations.	Four text-box summary layout.

11. Selected Visual Outputs for Case Study

The following visuals are screenshots from the Power BI dashboard and highlight selected outputs from the COVID-19 mortality analysis.

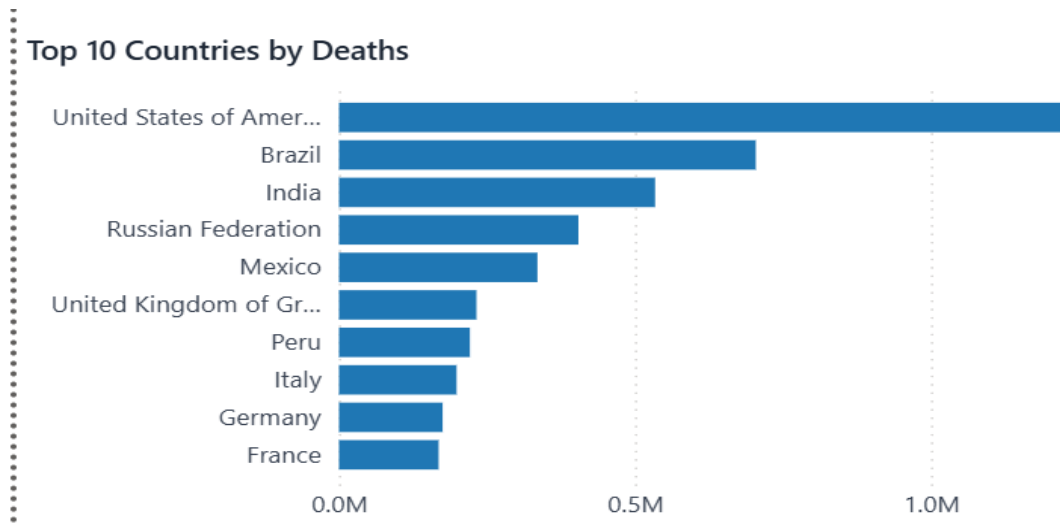


Figure 1: Top 10 countries by reported COVID-19 deaths.
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Global Weekly COVID-19 Deaths Trend

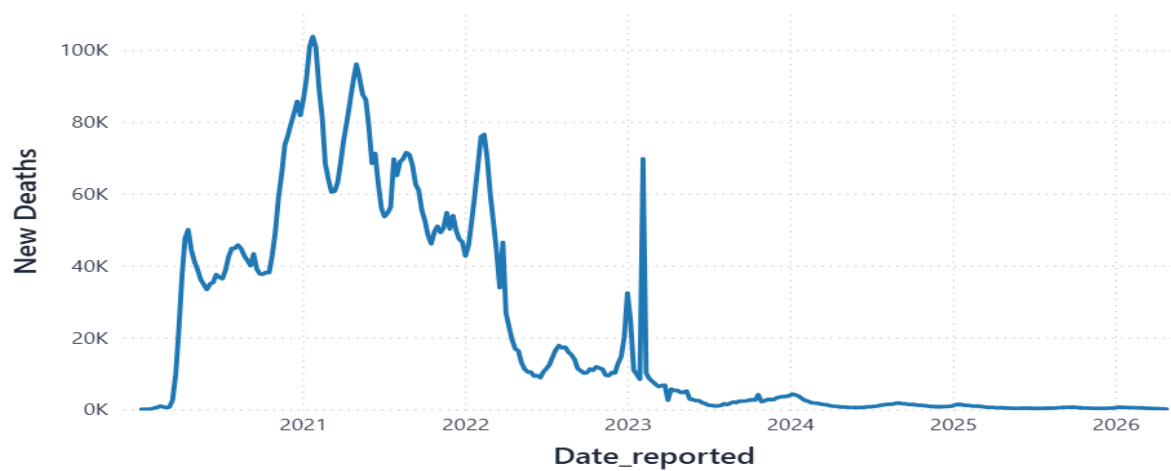


Figure 2: Global monthly reported COVID-19 deaths trend.

Zimbabwe Weekly COVID-19 Deaths Trend

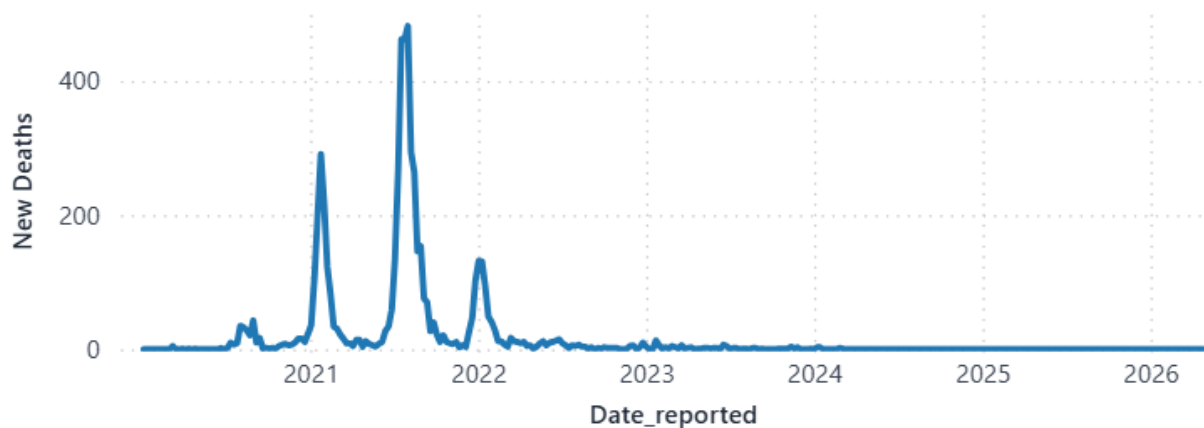


Figure 3: Zimbabwe reported COVID-19 deaths trend.

12. Key Findings

- COVID-19 mortality varied significantly across countries and regions, showing that country comparisons should consider both totals and context.
- The dataset records 779,213,618 reported cases and 7,114,800 reported deaths across 240 countries/locations over the reporting period analysed.
- Countries with large populations generally recorded high total deaths, but case fatality rate and deaths per population are more useful for fairer comparison.
- The United States of America, Brazil, India, Russian Federation and Mexico were among the countries with the highest reported COVID-19 deaths in the dataset.
- Zimbabwe recorded 266,436 reported cases and 5,740 reported deaths, producing a case fatality rate of approximately 2.15% within the dataset.
- Weekly or monthly death trends are useful for detecting periods of increased disease burden and possible pressure on health systems.

13. Public Health Interpretation

- Monitoring deaths over time helps health teams identify periods where mortality increased and where further investigation may be needed.
- Country-level dashboards support rapid comparison, but results must be interpreted alongside population size, testing coverage and reporting quality.
- Zimbabwe's focused page demonstrates how a national-level dashboard can support public health monitoring and reporting.
- Slicers for reporting year, WHO region and country allow users to interactively explore the data rather than relying on static tables.

14. Data Limitations

- Reported COVID-19 deaths may not represent all actual deaths because reporting practices differ between countries.
- Testing availability, reporting delays and changes in public health reporting rules can affect case and death counts.
- Case fatality rate is influenced by testing levels, healthcare access, age structure, reporting completeness and case definitions.
- Weekly reporting smooths daily variation, but it may hide short-term changes that happened within a reporting week.
- This dashboard is suitable for descriptive trend analysis and reporting, not for clinical diagnosis or a full epidemiological model.

15. Recommendations

- Use total deaths together with case fatality rate and population-adjusted indicators for more balanced comparison.
- Track weekly deaths over time to identify unusual increases early.
- Combine case and death data with vaccination, testing and hospital-capacity data for deeper analysis in future portfolio extensions.
- Improve completeness and consistency of public health reporting to support better decision-making.
- Use dashboards to communicate key health trends clearly to health managers, policy teams and public health stakeholders.

16. Project Outcome

This project demonstrates the ability to work with health data, prepare analysis-ready datasets, create meaningful calculations, build interactive dashboards and communicate findings in a clear public health reporting format.

It also shows practical use of Excel, SQL and Power BI to turn raw health data into decision-support insights. The project can be presented as the first major health data analytics portfolio project, with later projects added in health screening analytics, LMS engagement analytics, DHIS2-style reporting and hospital or facility performance analysis.